

Chapter 2

The Nature of Volatility Trading and Implied Volatility

Traders often hedge against periods of extreme market volatility (either to the upside or the downside) using options. Options are effectively financial insurance, and they are priced according to similar principles as other forms of insurance. Premiums increase or decrease according to the *perceived* risk of a given underlying (a result of supply and demand for those contracts), just as the cost of hurricane insurance increases or decreases depending on the perceived risk of hurricanes in a given area. To quantify the perceived risk in the market, traders use implied volatility (IV).

Implied volatility is the value of volatility that would make the current market price for an option be the fair price for that option in a given model, such as Black-Scholes.¹ When options prices *increase* (i.e., there is more demand for insurance), IV increases accordingly, and when options prices decrease, IV decreases. IV is, thus, a proxy for the *sentiment* of market risk as it relates to supply and demand for financial insurance. IV gives the perceived *magnitude* of expected price movements; it is not directional.²

[Table 2.1](#) gives a numerical example.

Table 2.1 Two underlyings with the same price and put contracts on each underlying with identical parameters (number of shares, put strike, contract duration). The contract prices differ, indicating that these two instruments have different implied volatilities.

45-day Put Contract	Underlying A	Underlying B
Underlying Price	\$101	\$101
Strike Price	\$100	\$100
Contract Price	\$10	\$5

The price of the put is around 10% of the stock price for underlying A and 5% of the stock price for underlying B. This suggests that there is more

perceived uncertainty associated with the price of underlying A compared to underlying B. Equivalently, this indicates that the anticipated magnitude of future moves in the underlying price is larger for underlying A compared to underlying B.

Demand for options tends to increase when the historical volatility of an underlying increases unexpectedly, particularly with large moves to the downside. This means that IV tends to be positively correlated with historical volatility and negatively correlated with price. However, there are exceptions to this rule, as IV is based on the perceived risk and not on historical risk directly. IV may increase due to factors that are not directly related to price movements, such as company-specific uncertainty (earnings reports, silly tweets from the CEO) or larger-scale macroeconomic uncertainty (political conflict, proposed legislative measures). This also means that volatility profiles vary significantly from instrument to instrument, which will be discussed more later in the chapter.

Similar to historical volatility, IV gives a one standard deviation range of annual returns for an instrument. Though historical volatility represents the realized *past volatility of returns*, IV is the approximation for *future volatility of returns* because it is based on how the market is using options to hedge against future price changes. While each option for an underlying has its own implied volatility, the “overall” IV of an asset is normally calculated from 30-day options and is a rough annualized volatility forecast.³

Example: An asset has a price of \$100 and an IV of 0.10 (10%). Therefore, the asset is expected to move about 10% to the upside or the downside by the end of the following year. This means the ending price will most likely be between \$90 and \$110.

The volatility forecast can also be scaled to approximate the expected price across days, weeks, months, or longer. The equations used to calculate the expected price ranges of an asset over some forecasting period are given below.⁴

$$1\sigma \text{ expected range (\%)} = IV \cdot \sqrt{\frac{\text{No. of Calendar Days}}{365}} \quad (2.1)$$

$$1\sigma \text{ expected range (\$)} = \text{Stock price} \cdot IV \cdot \sqrt{\frac{\text{No. of Calendar Days}}{365}} \quad (2.2)$$

These estimates of expected range will be used to formulate options strategies in future chapters. The time frame for the expected range is often scaled to match the contract duration. Most examples in this book will have a duration of 45 days to expiration (DTE) (or 33 trading days), so implied volatilities are typically multiplied by 0.35 to ensure forecasts match the duration of the contract.

The expected move cone is helpful to visualize this likely price range for an instrument according to market speculation. The width of the cone is calculated using [Equation \(2.2\)](#) and scales with the IV of the underlying. More specifically, the cones are wider in higher volatility environments and narrower when volatility is low and the expected range is tighter. Consider the expected move cones shown in [Figure 2.1](#), corresponding to the expected price ranges for SPY.

[Figure 2.1\(c\)](#) shows the realized price trajectory for SPY in December 2019, which stayed within its expected price range for the majority of the 45-day duration. Prices tend to stay within their expected range more often than not, and the assumptions of the Black-Scholes model can be used to develop a theoretical estimate for how often that should be.

Trading Volatility

An inconceivable number of factors affect prices in financial markets, which makes precisely forecasting price movements extremely difficult. Arguably, the most reliable way to form expectations around future price trends is using statistics from past price data and financial models. IV is derived from current options prices and the Black-Scholes options pricing

model, meaning that the Black-Scholes assumptions can be used to add statistical context to the expected price range. More specifically, one can infer the likelihood of a stock price remaining within its IV-derived price range because stock returns are assumed to be normally distributed. The one standard deviation range of the normal distribution encompasses 68.2% of event outcomes, so there is theoretically a 68.2% chance the price of an equity lands within its expected range. This probability can also be generalized over any timescale using [Equation \(2.1\)](#).

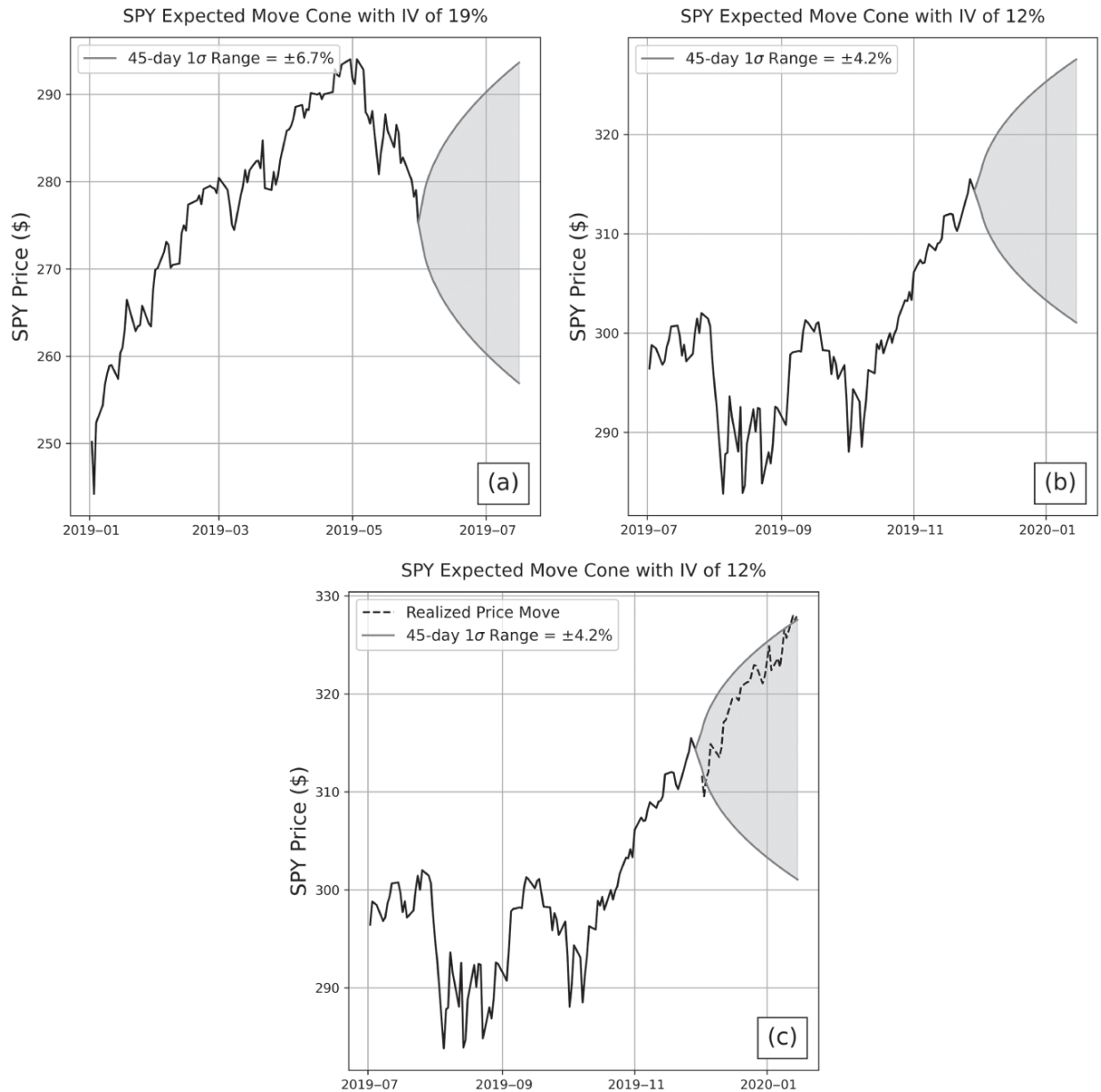


Figure 2.1 (a) The 45-day expected move cone for SPY in early 2019. The price of SPY was roughly \$275, and the IV was around 19%, corresponding to a 45-day expected price range of $\pm 6.7\%$ ([Equation \(2.1\)](#)) or $\pm \$18$ ([Equation \(2.2\)](#)). (b) The 45-day expected move cone for SPY when IV was 12%. (c) The same expected move cone as (b) with the realized price over 45 days.

Example: An asset has a price of \$100 and an IV of 0.10 (10%). The asset price is expected to remain between \$90 and \$110 by the end of the following year with 68% certainty. Equivalently, the asset price is expected to remain between \$96 and \$104 58 days from today with 68% certainty (calculated using [Equation \(2.2\)](#)).

However, historical data show that perceived uncertainty in the market (IV) tends to overstate the realized underlying price move more often than theory suggests. Though theory predicts that IV should overstate the realized move roughly only 68% of the time, market IV (estimated using the IV for SPY) overstated the realized move 87% of the time between 2016 and 2021. This means the price for SPY stayed within its expected price range more often than estimated. Realized moves were larger just 13% of the time, indicating that IV rarely understates the realized risk in the market. The *exact* degree to which IV tends to overstate realized volatility depends on the instrument. For example, consider the IV overstatement rates of the stocks and exchange-traded funds (ETFs) in [Table 2.2](#).

Table 2.2 IV overstatement of realized moves for six assets from 2016–2021. Assets include SPY (S&P 500 ETF), GLD (gold commodity ETF), SLV (silver commodity ETF), AAPL (Apple stock), GOOGL (Google stock), AMZN (Amazon stock).

Volatility Data (2016–2021)	
Asset	IV Overstatement Rate
SPY	87%
GLD	79%
SLV	89%
AAPL	70%
GOOGL	79%
AMZN	77%

Different assets are more or less prone to stay within their expected move range depending on their unique risk profile. Stocks are subject to single-company risk factors and tend to be more volatile. ETFs, which contain a variety of assets, are inherently diversified and tend to be less prone to dramatic price swings. For example, the S&P 500 includes Apple, but it

also includes around 499 other companies. This means that a tech-sector specific event will have a bigger impact on APPL compared to SPY. Commodities like gold and silver also tend to be less volatile than individual stocks, meaning they are less prone to spikes in IV and have more predictable returns. Although the IV overstatement rates differ between instruments, one can conclude that *fear* of large price moves is usually greater than realized price moves in the market. So, how exactly can options traders capitalize on this knowledge of IV and IV overstatement?

Let's revisit the example of hurricane insurance. The price for hurricane insurance is proportional to the expected cost of potential hurricane damage in the area. These prices are based on historical hurricane activity and forecasts of future events, which may underestimate, overestimate, or match the realized outcomes. People who *sell* hurricane insurance initially collect premiums, with the value depending on the perceived risk of home damage. During uneventful hurricane seasons, most policies go unused, and insurers keep the majority of premiums initially collected. In the unlikely event that hurricane damage is *significantly* worse than expected in an area dense with policyholders, insurers take very large losses. Insurance companies essentially make small, consistent profits the majority of the time while being exposed to large, infrequent losses.

Financial insurance carries a similar risk-reward trade-off as sellers make small, consistent profits most of the time but run the risk of large losses in extreme circumstances. IV yields an approximate price range forecast for a given underlying with 68% certainty. This means there is a 68% chance that the calls with strikes at the upper end of the expected range and puts with strikes at the lower end will both expire with no intrinsic value. For example, if traders sold one call and one put with strikes along the expected move cone, they would theoretically profit with 68% certainty. If the underlying price were to move unexpectedly to the upside or the downside, however, the traders may take substantial losses.

Unlike sellers of hurricane insurance, options sellers have more room to strategize and more control over their risk-reward profile. Premium sellers can choose when to sell insurance and how to construct contracts most likely to be profitable. Because IV is a proxy for the demand for options and the inflation of premium, it can be used to identify opportune times to sell insurance. Additionally, because IV can be used to estimate the most

likely price range for a specific asset, premium sellers can use IV to structure those positions so they likely expire worthless, like in the previous example. Options sellers (or short premium traders) have the long-term statistical advantage over options buyers, with the trade-off of exposure to unlikely, potentially significant losses. Because of that long-term statistical advantage, short premium trading is the focus of this book, with the next chapter detailing the mechanics of trading based on implied volatility.

The States of VIX

SPY is frequently used as a proxy for the broader market. It is also a baseline underlying for the short options strategies in this book because it is highly diversified across market sectors and has minimal idiosyncratic risk factors. The CBOE Volatility Index (VIX) is meant to track the annualized IV for SPY and is derived from 30-day index options. As SPY is a proxy for the broader market, the VIX, therefore, gauges the perceived risk of the broader market. For context, from 1990 to 2021, the VIX ranged from roughly 10 to a peak of just over 80 in March 2020 during the COVID-19 pandemic.⁵

Unlike equities, whose prices typically drift from their starting values over time, IV tends to revert back to a long-term value following a cyclic trend. This is because equities are used to estimate the perceived value of a company, sector, or commodity, but IV tracks the uncertainty sentiment of the market, which can only stay elevated for so long. During typical bull market conditions, the VIX hovers at a relatively low value at or below its average of 18.5. This is known as a lull state. When market uncertainty rapidly increases for whatever reason, often in response to large sudden price changes, the VIX expands and spikes far above its steady-state value. Once the market adjusts to the new volatility conditions or the volatile conditions dissipate, the VIX gradually contracts back to a lull state. To see an example of this cycle, refer to [Figure 2.2](#).

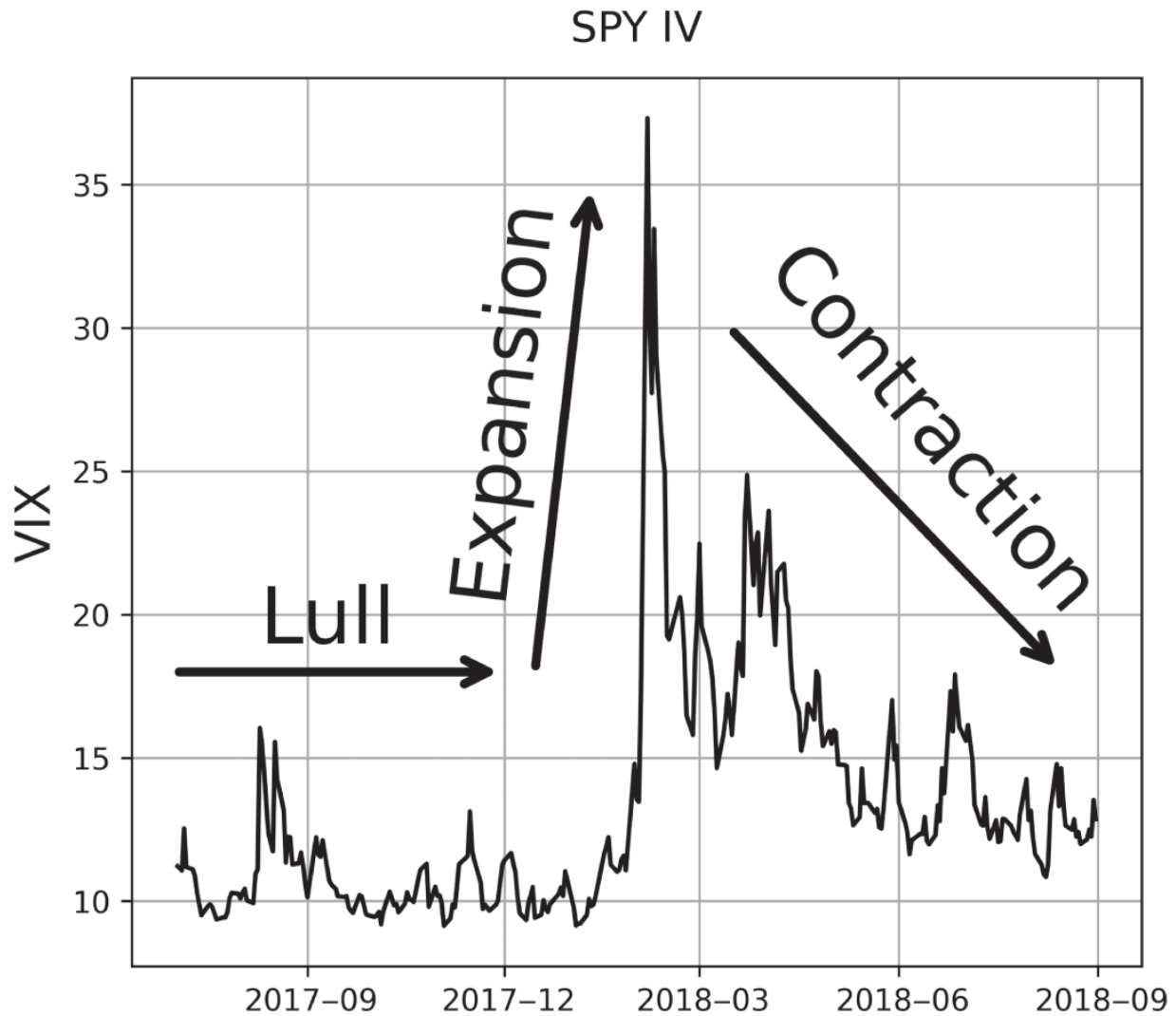


Figure 2.2 The three phases of the VIX, using data from early 2017 to late 2018.

When comparing how often the VIX is in each state, one finds the following approximate rates:

- Lull (70%): IV consistently remains below or near its long-term average. This state occurs when market prices trend upward gradually and market uncertainty is consistently low.
- Expansion (10%): IV expansion usually follows a prolonged lull period and is marked by expanding market uncertainty and typically large price moves in the underlying equity.

- Contraction (20%): IV contraction follows an expansion and is marked by a continued decline in IV. A contraction turns into a lull when IV reverts back to its long-term average.

Lull periods are most common and tend to be much longer than the average expansion or contraction period. Since 2000, the average lull period was more than three times the length of the average expansion or contraction. When expansions do happen, the higher the IV peak, the faster the VIX contracts. For example, according to data from 2005 to 2020, when the VIX contracted from 20 to 16 points (20% decrease), it took an average of 75.3 trading days to do so. However, when the VIX contracted from 70 to 56 points (also a 20% decrease), it only took an average of four trading days.

Spikes in the VIX are generally caused by unprecedented market or worldwide events. For example, the VIX reached over 80 in November 2008 during the peak of the worldwide financial crisis and hit its all-time high of 82.69 in March 2020 during the COVID-19 pandemic. The VIX peak of 2020 was especially unprecedented as the first major spike due to COVID-19 happened on February 28, 2020 when the VIX hit 40.11. This VIX high in 2020 had not been reached since February 2018, and it followed a 96-day lull. On March 16, 2020, the VIX hit 82.69, making the 2020 VIX expansion one of the most rapid ever recorded.

Though contraction periods tend to be longer than expansions but much shorter than lulls, fairly long contractions tend to follow major sell-offs or corrections. For example, the VIX contraction following the 2008 sell-off lasted well over a year, and the contraction following the 2020 sell-off lasted more than 10 months. This is normally because it takes time for the market (and specific subsectors) to revert to regular conditions following such broad macroeconomic shocks.

Premium sellers can potentially profit in any type of market, whether it be during volatility expansions (bearish), contractions (bullish/neutral), or lulls (neutral) if adopting an appropriate strategy for the volatility conditions. Generally, the most favorable trading state for selling premium is when IV contracts. This is because IV contracts when premium prices deflate, meaning that options traders who sold positions in high IV⁶ are able to buy identical positions back in low IV at a lower price, thus profiting from the

difference. Volatility expansions, on the other hand, have the potential to generate significant losses for short premium traders.

Volatility expansions tend to occur when there are large movements in the underlying price and uncertainty increases, causing options on that underlying to become more expensive. If traders sell premium during an expansion period once IV is *already elevated*, then the traders can capitalize on higher premium prices and the increased likelihood of a volatility contraction. However, if traders sell premium during a lull period, when the expected range is tight, and volatility *transitions* into an expansion period, then those traders will likely take large losses from the underlying price moving far outside the expected range. Additionally, to close their positions early, traders must buy back their options for more than they received in initial credit and incur a loss from the difference.

Short premium traders can profit in any type of market, but the risk of significant losses for short premium traders is highest when volatility is *low*. Unexpected transitions from a volatility lull to an expansion do not happen often, but when they do happen, they can be detrimental to an account. It is still necessary to trade during these low-IV periods because IV spends the majority of the time in this state, but risk management during this period is crucial. These risk management techniques will be outlined in the upcoming chapters.

This cyclic trend (lull, expansion, contraction, lull) is easily observable when looking at a relatively stable volatility index, such as the VIX. However, this trend, which we will describe as IV reversion, is present in some capacity for *all* IV signals.

IV Reversion

Certain types of signals tend to revert back to a long-term value following a significant divergence. Although this concept cannot be empirically proven or disproven, the reversion of IV is a core assumption in options trading.⁷ The reversion dynamics and the minimum IV level vary across instruments, but reversion is assumed to be present in *all* IV signals to some extent. To understand this, first consider the probability of large magnitude returns for

four assets with different risk profiles: SPY, GLD, AAPL, and AMZN. A comparison of these probabilities is shown in [Table 2.3](#).

Table 2.3 Rates that different assets experienced daily returns larger than 1%, 3%, and 5% in magnitude. For example, there is a 22% chance that SPY returns more than 1% or less than -1% in a single day (according to past data).

Probability of Surpassing Daily Returns Magnitude (2015–2021)			
Asset	> 1% Magnitude	> 3% Magnitude	> 5% Magnitude
SPY	22%	3%	0.8%
GLD	19%	1%	0.1%
AAPL	43%	9%	2%
AMZN	45%	10%	3%

Compared to assets like SPY and GLD, AMZN and AAPL are more volatile. These tech stocks experience large daily returns roughly three times as often as SPY and roughly 10 times as often as GLD. Each of these assets is subject to unique risk factors, but all are expected to have reverting IV signals nonetheless. [Figure 2.3](#) shows these volatility profiles graphically.

[Figure 2.3](#) demonstrates how IV has tended to revert back to a long-term baseline for each of the different assets, and it also demonstrates that elevated uncertainty is *unsustainable* in financial markets. Events may occur that spark fear in the market and drive up the demand for insurance, but as fear inevitably dissipates and the market adapts to the new conditions, IV deflates back down. This phenomenon has significant implications for short options traders. As stated in [Chapter 1](#), it is controversial whether directional price assumptions are statistically valid or not as trading according to pricing forecasts has never been proven to consistently outperform the market. IV is assumed to eventually revert down following inflations from its stable volatility state unlike asset prices, which drift from their initial value with time. The timescale for these contractions is unpredictable, but this nonetheless indicates some statistical validity to make downward directional assumptions about volatility once it is elevated.

[Figure 2.3](#) also shows how volatility profiles vary greatly across instruments. More volatile assets like Apple and Amazon stocks have higher IV averages, twice that of SPY and gold in this case, and experience expansion events more often. Single-company factors, such as quarterly earnings reports, pending mergers, acquisitions, and executive changes can all cause volatility spikes not seen in diversified assets and portfolios. However, this increased volatility also comes with higher credits and more volatility contraction opportunities for premium sellers.⁸ For an example of how the propensity for expansions and contractions differs between stocks with earnings and a diversified ETF, refer to [Figures 2.4\(a\)–\(c\)](#). Marked are the earnings report dates for each stock or the date when the company reported its quarterly profits (after-tax net income).

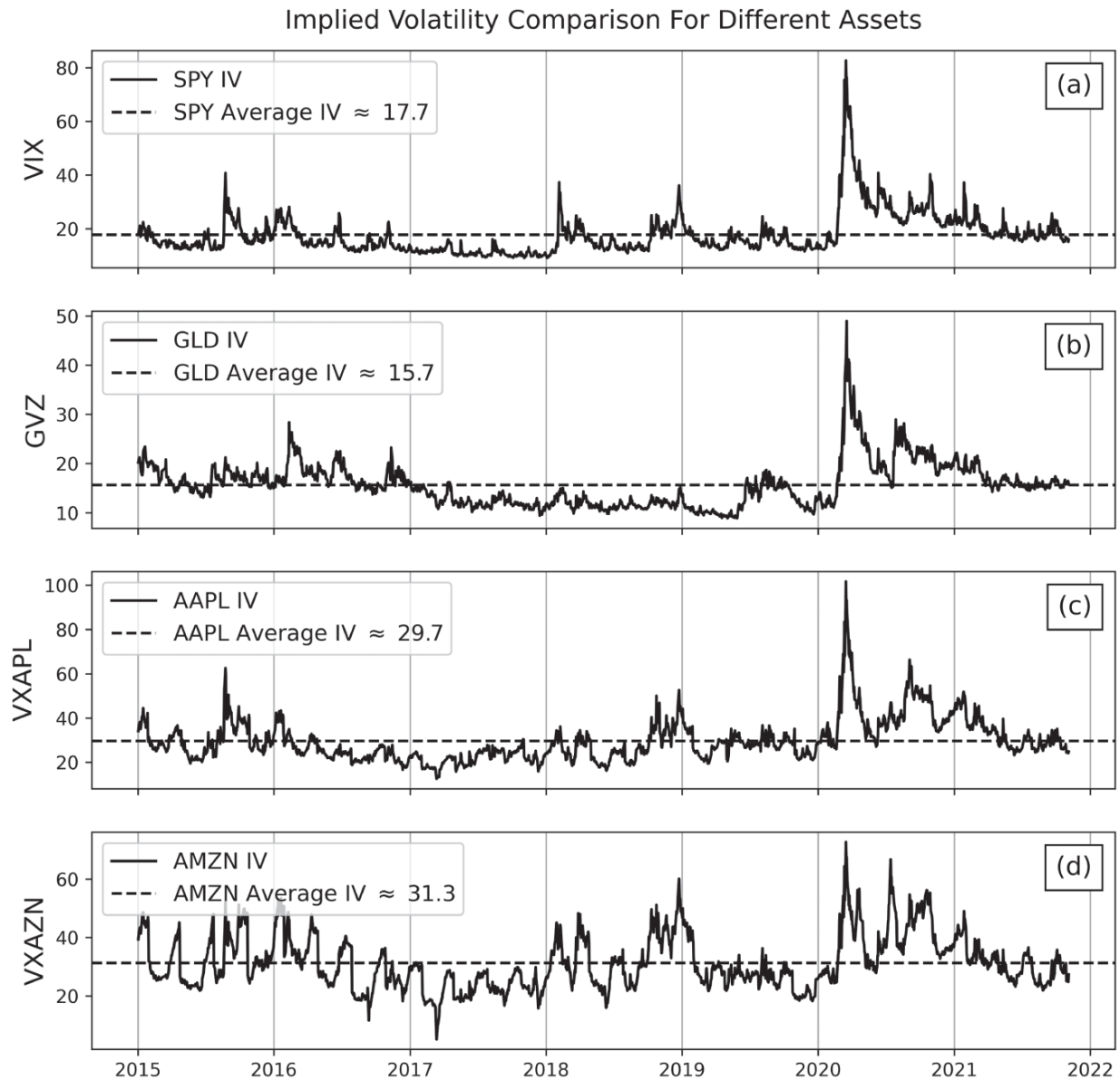


Figure 2.3 IV indexes for different assets with their respective averages (dashed) from 2015–2021. Assets include (a) SPY (S&P 500 ETF), (b) GLD (gold commodity ETF), (c) AAPL (Apple stock), and (d) AMZN (Amazon stock).

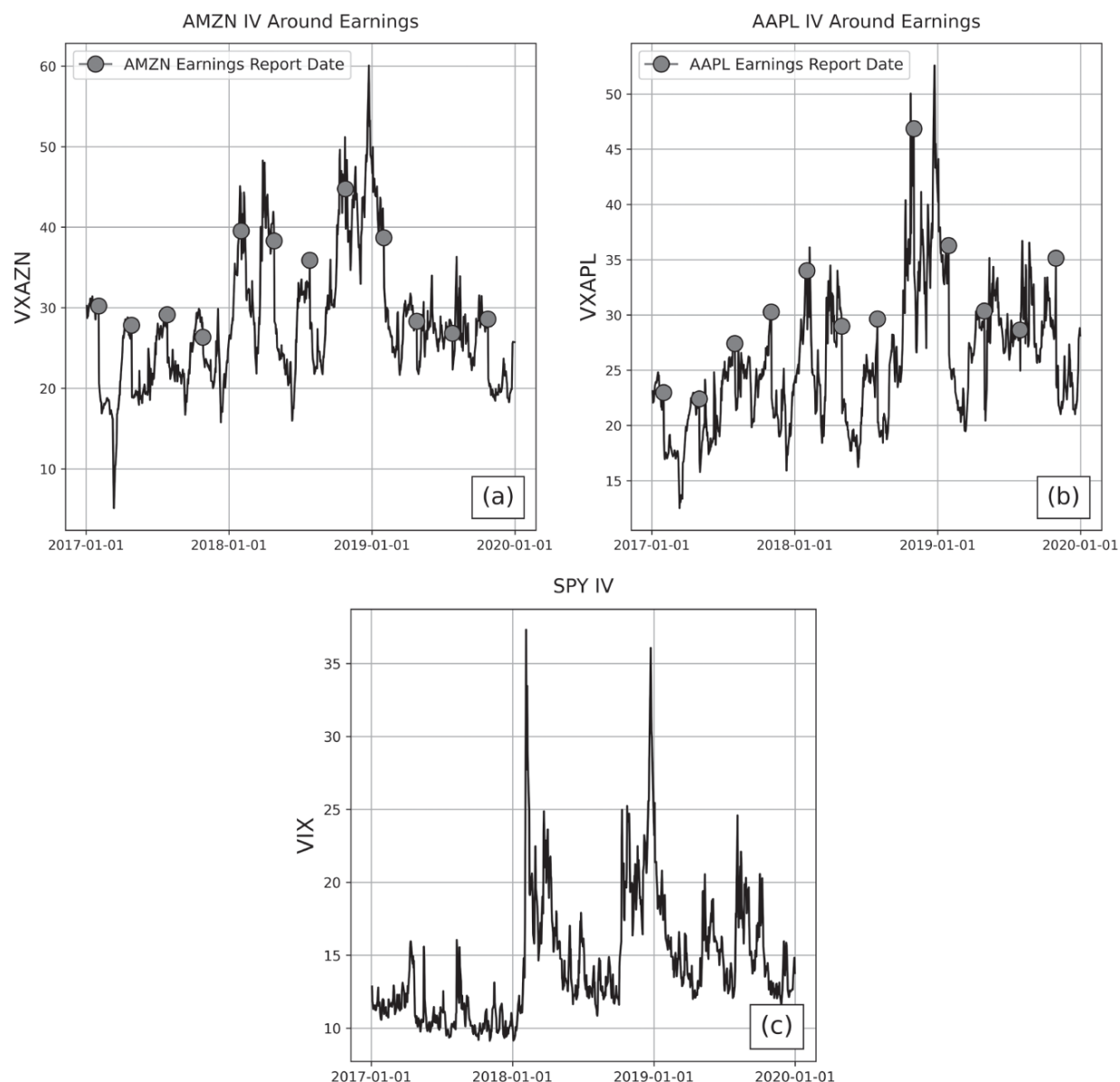


Figure 2.4 Implied volatility indexes for different equities from 2017–2020 with earnings dates marked (if applicable). Assets include (a) AMZN (Amazon stock), (b) AAPL (Apple stock), and (c) SPY (S&P 500 ETF).

With tech stocks like AMZN and AAPL, it's common for IV to increase sharply prior to earnings and contract almost immediately afterward. The previous graphs show that sharp IV expansions happen less frequently with a more diversified market ETF, such as SPY. These figures indicate that when SPY does experience a volatility expansion, it generally takes much longer to contract. From 2017 to 2020, the VIX only rose above 35 two times and, in both situations, took roughly half a month to contract down to

its original level. Meanwhile, volatility levels of AMZN and AAPL rose above 40 many times and even had a few spikes above 50, or in the case of AMZN, almost 60.

Takeaways

1. IV is a proxy for the sentiment of market risk derived from supply and demand. When options prices increase, IV increases; when options prices decrease, IV decreases. IV also gives the perceived magnitude of future movement, and it is not directional.
2. Demand for options tends to increase when the historical volatility of an underlying increases unexpectedly, particularly with large moves to the downside. IV tends to be positively correlated with historical volatility and negatively correlated with price, but it is ultimately based on the *perceived* market risk and not directly on price information.
3. IV can be used to estimate the expected price range of an instrument. IV gives a one standard deviation *expected* range because it is based on how the market is using options to hedge against future periods of volatility.
4. Because stock returns are assumed to be normally distributed, theoretically, there is a 68.2% chance the price of an equity lands within its expected range over a given time frame. However, historical data show that prices stay within their expected ranges more often than theoretically estimated. For example, market IV (estimated using the IV for SPY) overstated the realized move 87% of the time between 2016 and 2021.
5. Options sellers have the long-term statistical advantage over options buyers, with the trade-off of exposure to unlikely, potentially significant losses. Because IV is a proxy for the demand for options and the inflation of premium, it can be used to identify opportune times to sell insurance. Premium sellers can also use IV to structure positions so they are likely to expire worthless, the ideal outcome for the short position.

6. Volatility profiles differ significantly between assets, but all IV signals are assumed to revert back to some long-term value following significant diversions. Stated differently, IV tends to contract back to a long-term value following significant expansions from its lull volatility state. This phenomenon indicates that there is some degree of statistical validity when making downward directional assumptions about volatility once it's inflated.

Notes

- 1 Implied volatility (IV), like historical volatility, is a percentage and pertains to log returns. It is common to represent IV as either a decimal (0.X) or percentage (X%). An IV index, which is an instrument that tracks IV and will be introduced later in this chapter, is typically represented using points (X) but should be understood as a percentage (X%).
- 2 It is possible to get directional expected move information about an underlying by analyzing the IV across various strikes. This will be elaborated on more in the appendix.
- 3 IV yields a rough approximation for the expected price range, but this is not how the expected range is typically calculated on most trading platforms. Refer to the appendix for more information about how expected range is calculated more precisely. For the time being, we are using this simplified formula since it is most intuitive.
- 4 When ignoring the risk-free rate, the expected price range over T days for a stock with price S and volatility σ can be estimated by
$$Se^{\pm\sigma\sqrt{T/365}}$$
. The formula in [Equation \(2.2\)](#) is an approximation because, for small x values, $e^x \approx 1 + x$. This approximation becomes less valid when x is large, meaning this expected range calculation is less accurate when IV is high. This will be explored more in the appendix.
- 5 Note that volatility indices, such as the VIX, will be represented using points but are meant to be understood as a percentage. For example, a VIX of 30 corresponds to an annualized implied volatility of 30%.

- 6 It's important to note that the threshold for high IV is different for every asset because each instrument is subject to unique risk factors. Evaluating IV can be difficult because there is so much variability between assets, but there will be a more in-depth discussion of this in the following chapter.
- 7 The value that the signal reverts back to is roughly the long-term mode of the distribution, or the volatility that has occurred most often historically.
- 8 Such underlyings can be used for earnings plays, which will be discussed in a [Chapter 9](#).

OceanofPDF.com